
Math 31

Advanced Algebra

Math 31 Course at a Glance

Advanced Algebra

TEXTBOOK | OpenStax College Algebra 2e (Jay Abramson, 2021)

Review Content

REVIEW	SECTIONS	No mapped textbook section
LEARNING OBJECTIVES		
M31-REV-001	I can calculate a percentage, percent increase, or percent decrease.	
M31-REV-002	I can simplify an expression using exponent properties.	
M31-REV-003	I can use polynomial vocabulary to distinguish terms, factors, coefficients, degree, and polynomial structure.	
M31-REV-004	I can expand a product of two binomials.	

Linear Functions and Function Notation

REQUIRED	SECTIONS	3.1, 3.2, 3.3, 4.1, 4.2
LEARNING OBJECTIVES		
M31-LIN-001	I can interpret function notation as a statement about input and output.	
M31-LIN-002	I can determine whether a graph or table represents a function.	
M31-LIN-003	I can evaluate a function or solve for an input using a graph, table, or formula.	
M31-LIN-004	I can identify independent and dependent variables in a relationship.	
M31-LIN-005	I can calculate average rate of change from two points.	
M31-LIN-006	I can interpret interval notation and identify increasing or decreasing intervals.	
M31-LIN-007	I can determine total change from an average rate of change over an interval.	
M31-LIN-008	I can determine and interpret the units of an average rate of change.	
M31-LIN-009	I can identify a linear function from a table, graph, or formula.	
M31-LIN-010	I can interpret the slope and intercept of a linear model.	
M31-LIN-011	I can determine the intersection of two linear relationships graphically or algebraically.	
M31-LIN-012	I can write equations of horizontal, vertical, parallel, and perpendicular lines through a specified point.	

Function Behavior and Structure

REQUIRED	SECTIONS	3.1, 3.2, 3.3, 3.4, 3.5, 3.7
LEARNING OBJECTIVES		
M31-FUN-001	I can determine whether a relation is a function.	
M31-FUN-002	I can evaluate a function from a formula, graph, or table.	
M31-FUN-003	I can determine domain and range from a formula, graph, or table.	
M31-FUN-004	I can determine algebraic domain restrictions caused by denominators and even roots.	
M31-FUN-005	I can graph a piecewise function from its formula.	
M31-FUN-006	I can write a piecewise formula that represents a graph.	
M31-FUN-007	I can graph a transformed function using shifts, stretches, compressions, and reflections.	
M31-FUN-008	I can describe how a transformation affects domain, range, or asymptotes.	
M31-FUN-009	I can evaluate a composition from formulas, graphs, or tables.	
M31-FUN-010	I can write a formula for a composition of two functions.	
M31-FUN-011	I can find an inverse function algebraically.	
M31-FUN-012	I can evaluate and interpret an inverse from a graph, formula, or table.	
M31-FUN-013	I can determine and interpret the units of an inverse function.	
M31-FUN-014	I can explain how the domain and range of a function relate to those of its inverse.	
M31-FUN-015	I can determine the range of a function by analyzing the domain of its inverse.	
M31-FUN-016	I can verify algebraically that two functions are inverses.	
M31-FUN-017	I can interpret the meaning and units of a composition.	
M31-FUN-018	I can classify a graph or table as concave up or concave down using changes in average rate of change.	
M31-FUN-019	I can determine whether a function is invertible using the horizontal line test.	

Function Behavior and Structure

M31-FUN-020	I can graph an inverse function as a reflection across the line $y = x$.
--------------------	---

Math 31 Course at a Glance

Advanced Algebra

TEXTBOOK | OpenStax College Algebra 2e (Jay Abramson, 2021)

Quadratic Functions and Equations		Exponential Functions and Models		Logarithmic Functions and Models		Extension Content	
REQUIRED	SECTIONS 2.5, 5.1	REQUIRED	SECTIONS 6.1, 6.2, 6.7	REQUIRED	SECTIONS 6.3, 6.4, 6.5, 6.6, 6.7	EXTENSION	SECTIONS 3.6
LEARNING OBJECTIVES		LEARNING OBJECTIVES		LEARNING OBJECTIVES		LEARNING OBJECTIVES	
M31-QUAD-001	I can identify the vertex, intercepts, axis of symmetry, and concavity of a quadratic function.	M31-EXP-001	I can distinguish linear change from exponential change using differences, ratios, or percent change.	M31-LOG-001	I can convert between logarithmic and exponential forms.	M31-EXT-001	I can analyze an absolute-value function using transformations.
M31-QUAD-002	I can connect standard, factored, and vertex forms to features of a quadratic graph.	M31-EXP-002	I can identify initial value and growth or decay factor in an exponential model.	M31-LOG-002	I can evaluate a logarithm using its exponential meaning.	M31-EXT-002	I can determine even or odd symmetry from a graph or formula.
M31-QUAD-003	I can rewrite a quadratic function in a form that reveals a requested feature.	M31-EXP-003	I can write an exponential function from a table, graph, or context.	M31-LOG-003	I can expand or condense an expression using product, quotient, and power properties.	M31-EXT-003	I can interpret a contextual logarithmic scale.
M31-QUAD-004	I can solve a quadratic equation by factoring or using the quadratic formula.	M31-EXP-004	I can evaluate and interpret an exponential model.	M31-LOG-004	I can solve an exponential equation using logarithms.		
M31-QUAD-005	I can graph a quadratic function using its algebraic structure and key features.	M31-EXP-005	I can calculate growth under periodic compounding.	M31-LOG-005	I can solve a logarithmic equation using exponential form and check domain restrictions.		
M31-QUAD-006	I can construct a quadratic function from specified zeros, a vertex, or graphical information.	M31-EXP-006	I can calculate growth under continuous compounding.	M31-LOG-006	I can calculate and interpret doubling time or half-life.		
M31-QUAD-007	I can create and interpret a quadratic trajectory model.	M31-EXP-007	I can determine an exponential formula from two points or other sufficient data.	M31-LOG-007	I can determine domain, range, intercepts, and asymptotes of a logarithmic function.		
		M31-EXP-008	I can determine domain, range, intercepts, and asymptotes of an exponential function.	M31-LOG-008	I can explain how exponential and logarithmic functions undo one another.		
		M31-EXP-009	I can compare the growth rates of two exponential functions.	M31-LOG-009	I can solve a multi-step exponential or logarithmic equation.		
		M31-EXP-010	I can determine the percent growth or decay represented by a continuous exponential model.	M31-LOG-010	I can determine an intersection of exponential functions graphically or algebraically.		
				M31-LOG-011	I can rewrite a discrete exponential model as an equivalent continuous exponential model.		
				M31-LOG-012	I can solve and interpret an exponential or logarithmic modeling problem.		

Math 31 In-Depth Curriculum

Advanced Algebra

TEXTBOOK | OpenStax College Algebra 2e (Jay Abramson, 2021)

SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

Review Content

REVIEW

UNIT TEXTBOOK COVERAGE

No mapped textbook section

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M31-REV-001	I can calculate a percentage, percent increase, or percent decrease. TEXTBOOK SECTIONS No mapped textbook section
SUPPORTING SKILLS	
R	Calculate a percentage of a quantity or determine what percent one quantity is of another. (inherited from middle school) SK-NUM-PERCENT-CALCULATE · Reinforce
R	Calculate and interpret percent increase or decrease. (inherited from middle school) SK-NUM-PERCENT-CHANGE · Reinforce

M31-REV-002	I can simplify an expression using exponent properties. TEXTBOOK SECTIONS No mapped textbook section
SUPPORTING SKILLS	
R	Rewrite a negative exponent using a reciprocal. (first introduced in Math 21) SK-EXP-NEGATIVE · Reinforce
R	Apply the power rule for exponents. (first introduced in Math 12) SK-EXP-POWER · Reinforce
R	Apply the product rule for exponents. (first introduced in Math 12) SK-EXP-PRODUCT · Reinforce
R	Apply the quotient rule for exponents. (first introduced in Math 12) SK-EXP-QUOTIENT · Reinforce
R	Apply the zero-exponent rule. (first introduced in Math 21) SK-EXP-ZERO · Reinforce

M31-REV-003	I can use polynomial vocabulary to distinguish terms, factors, coefficients, degree, and polynomial structure. TEXTBOOK SECTIONS No mapped textbook section
SUPPORTING SKILLS	
R	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Reinforce

M31-REV-004	I can expand a product of two binomials. TEXTBOOK SECTIONS No mapped textbook section
SUPPORTING SKILLS	
A	Apply the distributive property in either direction. (inherited from middle school) SK-ALG-DISTRIBUTE · Apply
R	Multiply two binomials using distribution. (first introduced in Math 12) SK-POLY-BINOMIAL-MULTIPLY · Reinforce

Math 31 In-Depth Curriculum

Advanced Algebra

TEXTBOOK | OpenStax College Algebra 2e (Jay Abramson, 2021)

SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

Linear Functions and Function Notation

REQUIRED

UNIT TEXTBOOK COVERAGE

3.1 Functions and Function Notation; 3.2 Domain and Range; 3.3 Rates of Change and Behavior of Graphs; 4.1 Linear Functions; 4.2 Modeling with Linear Functions

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M31-LIN-001

I can interpret function notation as a statement about input and output.
TEXTBOOK SECTIONS 3.1 Functions and Function Notation

SUPPORTING SKILLS

D

Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12)
SK-FUNC-INPUT-OUTPUT · Deepen

A

Identify the input and output represented by function notation. (first introduced in Math 12)
SK-FUNC-NOTATION-READ · Apply

I

Use function notation to express and solve an input-output question.
SK-FUNC-NOTATION-SOLVE · Introduce

M31-LIN-002

I can determine whether a graph or table represents a function.
TEXTBOOK SECTIONS 3.1 Functions and Function Notation

SUPPORTING SKILLS

D

Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12)
SK-FUNC-INPUT-OUTPUT · Deepen

D

Apply the vertical line test to a graph. (first introduced in Math 12)
SK-FUNC-VERTICAL-LINE · Deepen

M31-LIN-003

I can evaluate a function or solve for an input using a graph, table, or formula.
TEXTBOOK SECTIONS 3.1 Functions and Function Notation

SUPPORTING SKILLS

A

Use inverse operations to maintain an equivalent equation. (first introduced in Math 12)
SK-ALG-INVERSE-OPERATIONS · Apply

D

Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12)
SK-FUNC-INPUT-OUTPUT · Deepen

A

Use function notation to express and solve an input-output question. (first introduced in Math 31)
SK-FUNC-NOTATION-SOLVE · Apply

D

Solve for an input associated with a specified output. (first introduced in Math 21)
SK-FUNC-SOLVE-INPUT · Deepen

M31-LIN-004

I can identify independent and dependent variables in a relationship.
TEXTBOOK SECTIONS 3.1 Functions and Function Notation

SUPPORTING SKILLS

I

Identify independent and dependent variables.
SK-FUNC-INDEPENDENT-DEPENDENT · Introduce

A

Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12)
SK-FUNC-INPUT-OUTPUT · Apply

M31-LIN-005

I can calculate average rate of change from two points.
TEXTBOOK SECTIONS 3.3 Rates of Change and Behavior of Graphs; 4.2 Modeling with Linear Functions

SUPPORTING SKILLS

D

Calculate average rate of change over an interval. (first introduced in Math 21)
SK-FUNC-AROC · Deepen

A

Calculate slope from two points. (first introduced in Math 12)
SK-LIN-SLOPE-POINTS · Apply

Math 31 In-Depth Curriculum

Advanced Algebra

TEXTBOOK | OpenStax College Algebra 2e (Jay Abramson, 2021)

SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

M31-LIN-006

I can interpret interval notation and identify increasing or decreasing intervals.

TEXTBOOK SECTIONS 3.2 Domain and Range; 3.3 Rates of Change and Behavior of Graphs

SUPPORTING SKILLS

I	Describe increasing or decreasing behavior over intervals. SK-FUNC-INTERVAL-BEHAVIOR · Introduce
D	Translate between inequality and interval notation. (first introduced in Math 12) SK-NOT-INTERVAL · Deepen

M31-LIN-007

I can determine total change from an average rate of change over an interval.

TEXTBOOK SECTIONS 3.3 Rates of Change and Behavior of Graphs; 4.2 Modeling with Linear Functions

SUPPORTING SKILLS

A	Calculate average rate of change over an interval. (first introduced in Math 21) SK-FUNC-AROC · Apply
I	Use average rate of change and interval length to calculate total change. SK-FUNC-AROC-TOTAL-CHANGE · Introduce
A	Add, subtract, multiply, and divide signed integers. (inherited from middle school) SK-NUM-INTEGGER-OPERATE · Apply

M31-LIN-008

I can determine and interpret the units of an average rate of change.

TEXTBOOK SECTIONS 3.3 Rates of Change and Behavior of Graphs; 4.2 Modeling with Linear Functions

SUPPORTING SKILLS

I	Determine rate units from input- and output-axis units. SK-FUNC-AROC-UNITS · Introduce
D	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Deepen

M31-LIN-009

I can identify a linear function from a table, graph, or formula.

TEXTBOOK SECTIONS 4.1 Linear Functions; 4.2 Modeling with Linear Functions

SUPPORTING SKILLS

I	Use constant differences or ratios to distinguish linear and exponential patterns. SK-EXP-DIFFERENCE-RATIO · Introduce
D	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Deepen
D	Use tabular patterns to predict graph shape and behavior. (first introduced in Math 21) SK-REP-TABLE-GRAPH · Deepen

M31-LIN-010

I can interpret the slope and intercept of a linear model.

TEXTBOOK SECTIONS 4.1 Linear Functions; 4.2 Modeling with Linear Functions

SUPPORTING SKILLS

A	Plot an intercept accurately on a coordinate plane. (first introduced in Math 12) SK-LIN-PLOT-INTERCEPT · Apply
A	Determine slope from a graph. (first introduced in Math 12) SK-LIN-SLOPE-GRAPH · Apply
D	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Deepen
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply

M31-LIN-011

I can determine the intersection of two linear relationships graphically or algebraically.

TEXTBOOK SECTIONS 4.2 Modeling with Linear Functions

SUPPORTING SKILLS

A	Check an ordered pair in both equations of a system. (first introduced in Math 12) SK-SYS-CHECK · Apply
A	Add equations to eliminate a variable. (first introduced in Math 12) SK-SYS-ELIMINATE · Apply
A	Locate and interpret the intersection of two lines. (first introduced in Math 12) SK-SYS-GRAPH-INTERSECTION · Apply
A	Substitute an equivalent expression for a variable. (first introduced in Math 12) SK-SYS-SUBSTITUTE · Apply

Math 31 In-Depth Curriculum

Advanced Algebra

TEXTBOOK | OpenStax College Algebra 2e (Jay Abramson, 2021)

SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

M31-LIN-012	I can write equations of horizontal, vertical, parallel, and perpendicular lines through a specified point. TEXTBOOK SECTIONS4.1 Linear Functions
SUPPORTING SKILLS	
A	Translate quantities and relationships into an equation. (first introduced in Math 12) SK-ALG-TRANSLATE-EQUATION · Apply
A	Convert among common forms of a linear equation. (first introduced in Math 12) SK-LIN-FORM-CONVERT · Apply
D	Recognize that distinct parallel lines have equal slopes. (first introduced in Math 12) SK-LIN-PARALLEL-SLOPE · Deepen
D	Determine the negative reciprocal of a nonzero slope. (first introduced in Math 12) SK-LIN-PERPENDICULAR-SLOPE · Deepen

Math 31 In-Depth Curriculum

Advanced Algebra

TEXTBOOK | OpenStax College Algebra 2e (Jay Abramson, 2021)

SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

Function Behavior and Structure

REQUIRED

UNIT TEXTBOOK COVERAGE

3.1 Functions and Function Notation; 3.2 Domain and Range; 3.3 Rates of Change and Behavior of Graphs; 3.4 Composition of Functions; 3.5 Transformation of Functions; 3.7 Inverse Functions

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M31-FUN-001	I can determine whether a relation is a function. TEXTBOOK SECTIONS 3.1 Functions and Function Notation
SUPPORTING SKILLS	
D	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Deepen
A	Apply the vertical line test to a graph. (first introduced in Math 12) SK-FUNC-VERTICAL-LINE · Apply

M31-FUN-002	I can evaluate a function from a formula, graph, or table. TEXTBOOK SECTIONS 3.1 Functions and Function Notation
SUPPORTING SKILLS	
A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
D	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Deepen
A	Solve for an input associated with a specified output. (first introduced in Math 21) SK-FUNC-SOLVE-INPUT · Apply

M31-FUN-003	I can determine domain and range from a formula, graph, or table. TEXTBOOK SECTIONS 3.2 Domain and Range
SUPPORTING SKILLS	
I	Determine domain restrictions from denominators, even roots, and logarithms. SK-FUNC-DOMAIN-FORMULA · Introduce
D	Read domain and range from a graph. (first introduced in Math 12) SK-FUNC-DOMAIN-RANGE-GRAPH · Deepen
D	Read domain and range from a table. (first introduced in Math 12) SK-FUNC-DOMAIN-RANGE-TABLE · Deepen

M31-FUN-004	I can determine algebraic domain restrictions caused by denominators and even roots. TEXTBOOK SECTIONS 3.2 Domain and Range
SUPPORTING SKILLS	
A	Factor a quadratic trinomial with a nonunit leading coefficient. (first introduced in Math 12) SK-FACTOR-TRINOMIAL-GENERAL · Apply
A	Factor a monic quadratic trinomial. (first introduced in Math 12) SK-FACTOR-TRINOMIAL-MONIC · Apply
D	Require an even-root radicand to be nonnegative when determining domain. (first introduced in Math 21) SK-FUNC-DOMAIN-EVEN-ROOT · Deepen
D	Determine domain restrictions from denominators, even roots, and logarithms. (first introduced in Math 31) SK-FUNC-DOMAIN-FORMULA · Deepen
A	Determine excluded values from an original denominator. (first introduced in Math 21) SK-RAT-RESTRICTIONS · Apply

M31-FUN-005	I can graph a piecewise function from its formula. TEXTBOOK SECTIONS No mapped textbook section
SUPPORTING SKILLS	
A	Describe increasing or decreasing behavior over intervals. (first introduced in Math 31) SK-FUNC-INTERVAL-BEHAVIOR · Apply
I	Select the correct rule of a piecewise function for a given input. SK-FUNC-PIECEWISE-EVALUATE · Introduce
I	Graph each rule on its stated domain with correct endpoints. SK-FUNC-PIECEWISE-GRAPH · Introduce

Math 31 In-Depth Curriculum

Advanced Algebra

TEXTBOOK | OpenStax College Algebra 2e (Jay Abramson, 2021)

SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

M31-FUN-006

I can write a piecewise formula that represents a graph.
TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

A	Describe increasing or decreasing behavior over intervals. (first introduced in Math 31) SK-FUNC-INTERVAL-BEHAVIOR · Apply
I	Translate a segmented graph into a piecewise formula. SK-FUNC-PIECEWISE-WRITE · Introduce
A	Convert among common forms of a linear equation. (first introduced in Math 12) SK-LIN-FORM-CONVERT · Apply

M31-FUN-007

I can graph a transformed function using shifts, stretches, compressions, and reflections.
TEXTBOOK SECTIONS 3.5 Transformation of Functions

SUPPORTING SKILLS

I	Interpret transformations applied to function inputs. SK-FUNC-TRANSFORM-HORIZONTAL · Introduce
I	Distinguish stretches from compressions and horizontal effects from vertical effects. SK-FUNC-TRANSFORM-SCALE · Introduce
I	Interpret transformations applied to function outputs. SK-FUNC-TRANSFORM-VERTICAL · Introduce
A	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Apply

M31-FUN-008

I can describe how a transformation affects domain, range, or asymptotes.
TEXTBOOK SECTIONS 3.5 Transformation of Functions

SUPPORTING SKILLS

I	Exchange domain and range between a function and its inverse. SK-FUNC-INVERSE-DOMAIN-RANGE · Introduce
D	Interpret transformations applied to function inputs. (first introduced in Math 31) SK-FUNC-TRANSFORM-HORIZONTAL · Deepen
D	Distinguish stretches from compressions and horizontal effects from vertical effects. (first introduced in Math 31) SK-FUNC-TRANSFORM-SCALE · Deepen
D	Interpret transformations applied to function outputs. (first introduced in Math 31) SK-FUNC-TRANSFORM-VERTICAL · Deepen

M31-FUN-009

I can evaluate a composition from formulas, graphs, or tables.
TEXTBOOK SECTIONS 3.4 Composition of Functions

SUPPORTING SKILLS

I	Substitute one function formula into another. SK-FUNC-COMPOSE-FORMULA · Introduce
I	Follow intermediate values to compose functions from tables. SK-FUNC-COMPOSE-TABLE · Introduce
A	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Apply

M31-FUN-010

I can write a formula for a composition of two functions.
TEXTBOOK SECTIONS 3.4 Composition of Functions

SUPPORTING SKILLS

A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
D	Substitute one function formula into another. (first introduced in Math 31) SK-FUNC-COMPOSE-FORMULA · Deepen

M31-FUN-011

I can find an inverse function algebraically.
TEXTBOOK SECTIONS 3.7 Inverse Functions

SUPPORTING SKILLS

A	Isolate a specified variable in an equation or formula. (first introduced in Math 12) SK-ALG-VARIABLE-ISOLATE · Apply
I	Exchange variables and solve to obtain an inverse formula. SK-FUNC-INVERSE-ALGEBRA · Introduce
I	Exchange input and output values when interpreting an inverse. SK-FUNC-INVERSE-SWAP · Introduce

Math 31 In-Depth Curriculum

Advanced Algebra

TEXTBOOK | OpenStax College Algebra 2e (Jay Abramson, 2021)

SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

M31-FUN-012

I can evaluate and interpret an inverse from a graph, formula, or table.

TEXTBOOK SECTIONS 3.7 Inverse Functions

SUPPORTING SKILLS

A	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Apply
A	Exchange variables and solve to obtain an inverse formula. (first introduced in Math 31) SK-FUNC-INVERSE-ALGEBRA · Apply
D	Exchange input and output values when interpreting an inverse. (first introduced in Math 31) SK-FUNC-INVERSE-SWAP · Deepen

M31-FUN-013

I can determine and interpret the units of an inverse function.

TEXTBOOK SECTIONS 3.7 Inverse Functions

SUPPORTING SKILLS

I	Reverse input and output units when interpreting an inverse. SK-FUNC-INVERSE-UNITS · Introduce
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply

M31-FUN-014

I can explain how the domain and range of a function relate to those of its inverse.

TEXTBOOK SECTIONS 3.7 Inverse Functions

SUPPORTING SKILLS

I	Exchange domain and range between a function and its inverse. SK-FUNC-INVERSE-DOMAIN-RANGE · Introduce
A	Exchange input and output values when interpreting an inverse. (first introduced in Math 31) SK-FUNC-INVERSE-SWAP · Apply

M31-FUN-015

I can determine the range of a function by analyzing the domain of its inverse.

TEXTBOOK SECTIONS 3.7 Inverse Functions

SUPPORTING SKILLS

A	Determine domain restrictions from denominators, even roots, and logarithms. (first introduced in Math 31) SK-FUNC-DOMAIN-FORMULA · Apply
D	Exchange domain and range between a function and its inverse. (first introduced in Math 31) SK-FUNC-INVERSE-DOMAIN-RANGE · Deepen

M31-FUN-016

I can verify algebraically that two functions are inverses.

TEXTBOOK SECTIONS 3.7 Inverse Functions

SUPPORTING SKILLS

A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
A	Substitute one function formula into another. (first introduced in Math 31) SK-FUNC-COMPOSE-FORMULA · Apply
I	Verify inverse functions using composition. SK-FUNC-INVERSE-VERIFY · Introduce

M31-FUN-017

I can interpret the meaning and units of a composition.

TEXTBOOK SECTIONS 3.4 Composition of Functions

SUPPORTING SKILLS

D	Substitute one function formula into another. (first introduced in Math 31) SK-FUNC-COMPOSE-FORMULA · Deepen
D	Follow intermediate values to compose functions from tables. (first introduced in Math 31) SK-FUNC-COMPOSE-TABLE · Deepen
A	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Apply
D	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Deepen

Math 31 In-Depth Curriculum

Advanced Algebra

TEXTBOOK | OpenStax College Algebra 2e (Jay Abramson, 2021)

SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

M31-FUN-018	I can classify a graph or table as concave up or concave down using changes in average rate of change. TEXTBOOK SECTIONS 3.3 Rates of Change and Behavior of Graphs
SUPPORTING SKILLS	
A	Calculate average rate of change over an interval. (first introduced in Math 21) SK-FUNC-AROC · Apply
I	Use successive average rates of change to classify concavity. SK-FUNC-CONCAVITY-AROC · Introduce
D	Use tabular patterns to predict graph shape and behavior. (first introduced in Math 21) SK-REP-TABLE-GRAPH · Deepen

M31-FUN-019	I can determine whether a function is invertible using the horizontal line test. TEXTBOOK SECTIONS 3.7 Inverse Functions
SUPPORTING SKILLS	
I	Apply the horizontal line test for invertibility. SK-FUNC-INVERSE-HLT · Introduce
I	Test whether a function is one-to-one on a domain. SK-FUNC-ONE-TO-ONE · Introduce

M31-FUN-020	I can graph an inverse function as a reflection across the line $y = x$. TEXTBOOK SECTIONS 3.7 Inverse Functions
SUPPORTING SKILLS	
A	Exchange domain and range between a function and its inverse. (first introduced in Math 31) SK-FUNC-INVERSE-DOMAIN-RANGE · Apply
D	Exchange input and output values when interpreting an inverse. (first introduced in Math 31) SK-FUNC-INVERSE-SWAP · Deepen
D	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Deepen

Math 31 In-Depth Curriculum

Advanced Algebra

TEXTBOOK | OpenStax College Algebra 2e (Jay Abramson, 2021)

SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

Quadratic Functions and Equations

REQUIRED

UNIT TEXTBOOK COVERAGE

2.5 Quadratic Equations; 5.1 Quadratic Functions

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M31-QUA D-001	I can identify the vertex, intercepts, axis of symmetry, and concavity of a quadratic function. TEXTBOOK SECTIONS 5.1 Quadratic Functions
SUPPORTING SKILLS	
A	Use successive average rates of change to classify concavity. (first introduced in Math 31) SK-FUNC-CONCAVITY-AROC · Apply
A	Identify standard, factored, and vertex forms. (first introduced in Math 21) SK-QUAD-FORM-IDENTIFY · Apply
D	Determine x- and y-intercepts of a quadratic function. (first introduced in Math 21) SK-QUAD-INTERCEPTS · Deepen
D	Determine the vertex and axis of symmetry. (first introduced in Math 21) SK-QUAD-VERTEX · Deepen

M31-QUA D-002	I can connect standard, factored, and vertex forms to features of a quadratic graph. TEXTBOOK SECTIONS 5.1 Quadratic Functions
SUPPORTING SKILLS	
D	Identify standard, factored, and vertex forms. (first introduced in Math 21) SK-QUAD-FORM-IDENTIFY · Deepen
A	Determine x- and y-intercepts of a quadratic function. (first introduced in Math 21) SK-QUAD-INTERCEPTS · Apply
A	Determine the vertex and axis of symmetry. (first introduced in Math 21) SK-QUAD-VERTEX · Apply
D	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Deepen

M31-QUA D-003	I can rewrite a quadratic function in a form that reveals a requested feature. TEXTBOOK SECTIONS 5.1 Quadratic Functions
SUPPORTING SKILLS	
A	Factor a quadratic trinomial with a nonunit leading coefficient. (first introduced in Math 12) SK-FACTOR-TRINOMIAL-GENERAL · Apply
A	Factor a monic quadratic trinomial. (first introduced in Math 12) SK-FACTOR-TRINOMIAL-MONIC · Apply
D	Create a perfect-square trinomial by completing the square. (first introduced in Math 21) SK-QUAD-COMPLETE-SQUARE · Deepen
D	Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method. (first introduced in Math 21) SK-QUAD-FORM-CONVERT · Deepen

M31-QUA D-004	I can solve a quadratic equation by factoring or using the quadratic formula. TEXTBOOK SECTIONS 2.5 Quadratic Equations; 5.1 Quadratic Functions
SUPPORTING SKILLS	
A	Apply the zero-product property. (first introduced in Math 21) SK-FACTOR-ZERO-PRODUCT · Apply
A	Use the discriminant to predict the number and type of solutions. (first introduced in Math 21) SK-QUAD-DISCRIMINANT · Apply
A	Substitute coefficients accurately into the quadratic formula. (first introduced in Math 21) SK-QUAD-FORMULA · Apply
D	Select an efficient quadratic-solving method from equation structure. (first introduced in Math 21) SK-QUAD-METHOD-SELECT · Deepen

Math 31 In-Depth Curriculum

Advanced Algebra

TEXTBOOK | OpenStax College Algebra 2e (Jay Abramson, 2021)

SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

M31-QUA D-005	I can graph a quadratic function using its algebraic structure and key features. TEXTBOOK SECTIONS5.1 Quadratic Functions
SUPPORTING SKILLS	
A	Identify standard, factored, and vertex forms. (first introduced in Math 21) SK-QUAD-FORM-IDENTIFY · Apply
D	Determine x- and y-intercepts of a quadratic function. (first introduced in Math 21) SK-QUAD-INTERCEPTS · Deepen
D	Determine the vertex and axis of symmetry. (first introduced in Math 21) SK-QUAD-VERTEX · Deepen
D	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Deepen

M31-QUA D-006	I can construct a quadratic function from specified zeros, a vertex, or graphical information. TEXTBOOK SECTIONS5.1 Quadratic Functions
SUPPORTING SKILLS	
A	Translate quantities and relationships into an equation. (first introduced in Math 12) SK-ALG-TRANSLATE-EQUATION · Apply
I	Build a polynomial from zeros, multiplicities, and a point. SK-POLY-CONSTRUCT · Introduce
A	Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method. (first introduced in Math 21) SK-QUAD-FORM-CONVERT · Apply

M31-QUA D-007	I can create and interpret a quadratic trajectory model. TEXTBOOK SECTIONS5.1 Quadratic Functions
SUPPORTING SKILLS	
D	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Deepen
A	Restrict a model to meaningful contextual inputs. (first introduced in Math 12) SK-MODEL-VALID-DOMAIN · Apply
D	Interpret vertex and zeros in an application. (first introduced in Math 21) SK-QUAD-MODEL-FEATURES · Deepen
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply

Math 31 In-Depth Curriculum

Advanced Algebra

TEXTBOOK | OpenStax College Algebra 2e (Jay Abramson, 2021)

SKILL PROGRESSION



Introduce



Reinforce



Deepen



Apply

Exponential Functions and Models

REQUIRED

UNIT TEXTBOOK COVERAGE

6.1 Exponential Functions; 6.2 Graphs of Exponential Functions; 6.7 Exponential and Logarithmic Models

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M31-EXP-001	I can distinguish linear change from exponential change using differences, ratios, or percent change. TEXTBOOK SECTIONS 6.1 Exponential Functions
SUPPORTING SKILLS	
D	Use constant differences or ratios to distinguish linear and exponential patterns. (first introduced in Math 31) SK-EXP-DIFFERENCE-RATIO · Deepen
D	Compare functions shown in different representations. (first introduced in Math 21) SK-FUNC-COMPARE-REP · Deepen
A	Calculate and interpret percent increase or decrease. (inherited from middle school) SK-NUM-PERCENT-CHANGE · Apply

M31-EXP-002	I can identify initial value and growth or decay factor in an exponential model. TEXTBOOK SECTIONS 6.1 Exponential Functions; 6.2 Graphs of Exponential Functions
SUPPORTING SKILLS	
I	Identify initial amount, rate, and time interval in context. SK-EXP-MODEL-CONTEXT · Introduce
I	Convert a percent increase or decrease to a growth or decay factor. SK-EXP-PERCENT-FACTOR · Introduce
D	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Deepen

M31-EXP-003	I can write an exponential function from a table, graph, or context. TEXTBOOK SECTIONS 6.1 Exponential Functions
SUPPORTING SKILLS	
D	Identify initial amount, rate, and time interval in context. (first introduced in Math 31) SK-EXP-MODEL-CONTEXT · Deepen
I	Determine an exponential model from tabular values. SK-EXP-MODEL-TABLE · Introduce
I	Solve for an exponential model using two points. SK-EXP-MODEL-TWO-POINTS · Introduce
D	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Deepen

M31-EXP-004	I can evaluate and interpret an exponential model. TEXTBOOK SECTIONS 6.1 Exponential Functions; 6.2 Graphs of Exponential Functions; 6.7 Exponential and Logarithmic Models
SUPPORTING SKILLS	
D	Identify initial amount, rate, and time interval in context. (first introduced in Math 31) SK-EXP-MODEL-CONTEXT · Deepen
A	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Apply
D	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Deepen
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply

M31-EXP-005	I can calculate growth under periodic compounding. TEXTBOOK SECTIONS 6.1 Exponential Functions; 6.7 Exponential and Logarithmic Models
SUPPORTING SKILLS	
I	Substitute correctly into a periodic-compounding model. SK-EXP-COMPOUND-PERIODIC · Introduce
A	Convert a percent increase or decrease to a growth or decay factor. (first introduced in Math 31) SK-EXP-PERCENT-FACTOR · Apply
A	Calculate a percentage of a quantity or determine what percent one quantity is of another. (inherited from middle school) SK-NUM-PERCENT-CALCULATE · Apply

Math 31 In-Depth Curriculum

Advanced Algebra

TEXTBOOK | OpenStax College Algebra 2e (Jay Abramson, 2021)

SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

M31-EXP-006

I can calculate growth under continuous compounding.
TEXTBOOK SECTIONS 6.1 Exponential Functions; 6.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

I

Substitute correctly into a continuous-compounding model.
SK-EXP-COMPOUND-CONTINUOUS · Introduce

A

Identify initial amount, rate, and time interval in context. (first introduced in Math 31)
SK-EXP-MODEL-CONTEXT · Apply

M31-EXP-007

I can determine an exponential formula from two points or other sufficient data.
TEXTBOOK SECTIONS 6.1 Exponential Functions

SUPPORTING SKILLS

A

Isolate a specified variable in an equation or formula. (first introduced in Math 12)
SK-ALG-VARIABLE-ISOLATE · Apply

D

Solve for an exponential model using two points. (first introduced in Math 31)
SK-EXP-MODEL-TWO-POINTS · Deepen

A

Interpret numerator and denominator roles in a rational exponent. (first introduced in Math 21)
SK-EXP-RATIONAL · Apply

M31-EXP-008

I can determine domain, range, intercepts, and asymptotes of an exponential function.
TEXTBOOK SECTIONS 6.2 Graphs of Exponential Functions

SUPPORTING SKILLS

I

Identify horizontal asymptotic behavior of an exponential graph.
SK-EXP-ASYMPTOTE · Introduce

D

Determine domain restrictions from denominators, even roots, and logarithms. (first introduced in Math 31)
SK-FUNC-DOMAIN-FORMULA · Deepen

D

Connect parameters in a formula to features of its graph. (first introduced in Math 21)
SK-REP-FORMULA-GRAPH · Deepen

M31-EXP-009

I can compare the growth rates of two exponential functions.
TEXTBOOK SECTIONS 6.2 Graphs of Exponential Functions

SUPPORTING SKILLS

I

Compare exponential growth rates expressed in different forms.
SK-EXP-COMPARE-GROWTH · Introduce

D

Compare functions shown in different representations. (first introduced in Math 21)
SK-FUNC-COMPARE-REP · Deepen

M31-EXP-010

I can determine the percent growth or decay represented by a continuous exponential model.
TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

I

Convert between equivalent discrete- and continuous-growth forms.
SK-EXP-DISCRETE-CONTINUOUS · Introduce

D

Convert a percent increase or decrease to a growth or decay factor. (first introduced in Math 31)
SK-EXP-PERCENT-FACTOR · Deepen

A

Calculate and interpret percent increase or decrease. (inherited from middle school)
SK-NUM-PERCENT-CHANGE · Apply

Math 31 In-Depth Curriculum

Advanced Algebra

TEXTBOOK | OpenStax College Algebra 2e (Jay Abramson, 2021)

SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

Logarithmic Functions and Models

REQUIRED

UNIT TEXTBOOK COVERAGE

6.3 Logarithmic Functions; 6.4 Graphs of Logarithmic Functions; 6.5 Logarithmic Properties; 6.6 Exponential and Logarithmic Equations; 6.7 Exponential and Logarithmic Models

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M31-LOG-001

I can convert between logarithmic and exponential forms.
TEXTBOOK SECTIONS 6.3 Logarithmic Functions

SUPPORTING SKILLS

I	Convert between logarithmic and exponential forms. SK-LOG-CONVERT · Introduce
I	Interpret a logarithm as an exponent. SK-LOG-DEFINITION · Introduce

M31-LOG-002

I can evaluate a logarithm using its exponential meaning.
TEXTBOOK SECTIONS 6.3 Logarithmic Functions

SUPPORTING SKILLS

I	Rewrite exponential expressions using a common base when possible. SK-EXP-REWRITE-BASE · Introduce
D	Interpret a logarithm as an exponent. (first introduced in Math 31) SK-LOG-DEFINITION · Deepen

M31-LOG-003

I can expand or condense an expression using product, quotient, and power properties.
TEXTBOOK SECTIONS 6.5 Logarithmic Properties

SUPPORTING SKILLS

A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
I	Apply the logarithm power property in either direction. SK-LOG-POWER · Introduce
I	Apply the logarithm product property in either direction. SK-LOG-PRODUCT · Introduce
I	Apply the logarithm quotient property in either direction. SK-LOG-QUOTIENT · Introduce

M31-LOG-004

I can solve an exponential equation using logarithms.
TEXTBOOK SECTIONS 6.6 Exponential and Logarithmic Equations

SUPPORTING SKILLS

A	Isolate a specified variable in an equation or formula. (first introduced in Math 12) SK-ALG-VARIABLE-ISOLATE · Apply
A	Apply the logarithm power property in either direction. (first introduced in Math 31) SK-LOG-POWER · Apply
I	Isolate an exponential expression and apply a logarithm. SK-LOG-SOLVE-EXP · Introduce

M31-LOG-005

I can solve a logarithmic equation using exponential form and check domain restrictions.
TEXTBOOK SECTIONS 6.6 Exponential and Logarithmic Equations

SUPPORTING SKILLS

A	Convert between logarithmic and exponential forms. (first introduced in Math 31) SK-LOG-CONVERT · Apply
I	Reject solutions that make a logarithm argument nonpositive. SK-LOG-DOMAIN-CHECK · Introduce
I	Combine or isolate logarithms before converting to exponential form. SK-LOG-SOLVE-LOG · Introduce

M31-LOG-006

I can calculate and interpret doubling time or half-life.
TEXTBOOK SECTIONS 6.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

I	Translate a doubling-time or half-life question into an exponential equation. SK-LOG-DOUBLING-HALF · Introduce
A	Isolate an exponential expression and apply a logarithm. (first introduced in Math 31) SK-LOG-SOLVE-EXP · Apply
D	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Deepen
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply

Math 31 In-Depth Curriculum

Advanced Algebra

TEXTBOOK | OpenStax College Algebra 2e (Jay Abramson, 2021)

SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

M31-LOG-007

I can determine domain, range, intercepts, and asymptotes of a logarithmic function.
TEXTBOOK SECTIONS 6.4 Graphs of Logarithmic Functions

SUPPORTING SKILLS

D	Determine domain restrictions from denominators, even roots, and logarithms. (first introduced in Math 31) SK-FUNC-DOMAIN-FORMULA · Deepen
I	Identify vertical asymptotic behavior of a logarithmic graph. SK-LOG-ASYMPTOTE · Introduce
A	Reject solutions that make a logarithm argument nonpositive. (first introduced in Math 31) SK-LOG-DOMAIN-CHECK · Apply
D	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Deepen

M31-LOG-008

I can explain how exponential and logarithmic functions undo one another.
TEXTBOOK SECTIONS 6.3 Logarithmic Functions; 6.4 Graphs of Logarithmic Functions

SUPPORTING SKILLS

A	Exchange input and output values when interpreting an inverse. (first introduced in Math 31) SK-FUNC-INVERSE-SWAP · Apply
A	Verify inverse functions using composition. (first introduced in Math 31) SK-FUNC-INVERSE-VERIFY · Apply
D	Interpret a logarithm as an exponent. (first introduced in Math 31) SK-LOG-DEFINITION · Deepen

M31-LOG-009

I can solve a multi-step exponential or logarithmic equation.
TEXTBOOK SECTIONS 6.6 Exponential and Logarithmic Equations

SUPPORTING SKILLS

A	Reject solutions that make a logarithm argument nonpositive. (first introduced in Math 31) SK-LOG-DOMAIN-CHECK · Apply
A	Apply the logarithm power property in either direction. (first introduced in Math 31) SK-LOG-POWER · Apply
A	Apply the logarithm product property in either direction. (first introduced in Math 31) SK-LOG-PRODUCT · Apply
A	Apply the logarithm quotient property in either direction. (first introduced in Math 31) SK-LOG-QUOTIENT · Apply
D	Isolate an exponential expression and apply a logarithm. (first introduced in Math 31) SK-LOG-SOLVE-EXP · Deepen
D	Combine or isolate logarithms before converting to exponential form. (first introduced in Math 31) SK-LOG-SOLVE-LOG · Deepen

M31-LOG-010

I can determine an intersection of exponential functions graphically or algebraically.
TEXTBOOK SECTIONS 6.6 Exponential and Logarithmic Equations

SUPPORTING SKILLS

A	Compare exponential growth rates expressed in different forms. (first introduced in Math 31) SK-EXP-COMPARE-GROWTH · Apply
D	Isolate an exponential expression and apply a logarithm. (first introduced in Math 31) SK-LOG-SOLVE-EXP · Deepen
A	Check an ordered pair in both equations of a system. (first introduced in Math 12) SK-SYS-CHECK · Apply
A	Locate and interpret the intersection of two lines. (first introduced in Math 12) SK-SYS-GRAPH-INTERSECTION · Apply

Math 31 In-Depth Curriculum

Advanced Algebra

TEXTBOOK | OpenStax College Algebra 2e (Jay Abramson, 2021)

SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

M31-LOG-011	I can rewrite a discrete exponential model as an equivalent continuous exponential model. TEXTBOOK SECTIONS No mapped textbook section
SUPPORTING SKILLS	
D	Convert between equivalent discrete- and continuous-growth forms. (first introduced in Math 31) SK-EXP-DISCRETE-CONTINUOUS · Deepen
A	Convert a percent increase or decrease to a growth or decay factor. (first introduced in Math 31) SK-EXP-PERCENT-FACTOR · Apply
A	Isolate an exponential expression and apply a logarithm. (first introduced in Math 31) SK-LOG-SOLVE-EXP · Apply

M31-LOG-012	I can solve and interpret an exponential or logarithmic modeling problem. TEXTBOOK SECTIONS 6.7 Exponential and Logarithmic Models
SUPPORTING SKILLS	
D	Identify initial amount, rate, and time interval in context. (first introduced in Math 31) SK-EXP-MODEL-CONTEXT · Deepen
A	Isolate an exponential expression and apply a logarithm. (first introduced in Math 31) SK-LOG-SOLVE-EXP · Apply
A	Combine or isolate logarithms before converting to exponential form. (first introduced in Math 31) SK-LOG-SOLVE-LOG · Apply
D	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Deepen
A	Restrict a model to meaningful contextual inputs. (first introduced in Math 12) SK-MODEL-VALID-DOMAIN · Apply
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply

Math 31 In-Depth Curriculum

Advanced Algebra

TEXTBOOK | OpenStax College Algebra 2e (Jay Abramson, 2021)

SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

Extension Content

EXTENSION

UNIT TEXTBOOK COVERAGE

3.6 Absolute Value Functions

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M31-EXT-001	I can analyze an absolute-value function using transformations. TEXTBOOK SECTIONS 3.6 Absolute Value Functions
SUPPORTING SKILLS	
A	Interpret transformations applied to function inputs. (first introduced in Math 31) SK-FUNC-TRANSFORM-HORIZONTAL · Apply
A	Distinguish stretches from compressions and horizontal effects from vertical effects. (first introduced in Math 31) SK-FUNC-TRANSFORM-SCALE · Apply
A	Interpret transformations applied to function outputs. (first introduced in Math 31) SK-FUNC-TRANSFORM-VERTICAL · Apply
A	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Apply

M31-EXT-002	I can determine even or odd symmetry from a graph or formula. TEXTBOOK SECTIONS No mapped textbook section
SUPPORTING SKILLS	
A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
I	Test a formula or graph for even or odd symmetry. SK-FUNC-SYMMETRY · Introduce

M31-EXT-003	I can interpret a contextual logarithmic scale. TEXTBOOK SECTIONS No mapped textbook section
SUPPORTING SKILLS	
A	Interpret a logarithm as an exponent. (first introduced in Math 31) SK-LOG-DEFINITION · Apply
I	Interpret equal intervals on a logarithmic scale as equal multiplicative changes. SK-LOG-SCALE-INTERPRET · Introduce
A	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Apply